

Algebraic Fractions (Rational Expressions)

Outcomes:

- Students will be able to simplify, add, subtract, multiply, divide, and solve equations involving algebraic fractions.

Algebraic Fractions:

- are fractions that have algebraic expressions in the numerator, denominator or both.
- can be polynomials containing variables, coefficients and constants
- just like numerical fractions, the denominator cannot be zero

Simplifying Algebraic Fractions

To simplify algebraic fractions we reduce the fraction to its simplest form by factoring and canceling out common terms in the numerator and the denominator.

Worked Examples:

$\frac{27xy^3}{9x^2y}$	Method:
$= \frac{3 \cdot 3 \cdot 3 \cdot x \cdot y \cdot y \cdot y}{3 \cdot 3 \cdot x \cdot x \cdot y}$	prime factorize the numerator and the denominator
$= \frac{3 \cdot y \cdot y}{x}$	cancel out common factors
$= \frac{3y^2}{x}$	simplify

$\frac{14x^4y^5}{21x^2y^8}$	Method:
$= \frac{7 \cdot 2 \cdot x \cdot x \cdot x \cdot x \cdot y \cdot y \cdot y \cdot y \cdot y}{7 \cdot 3 \cdot x \cdot x \cdot y \cdot y \cdot y \cdot y \cdot y \cdot y \cdot y}$	prime factorize the numerator and the denominator
$= \frac{2 \cdot x \cdot x}{3 \cdot y \cdot y \cdot y}$	cancel out common factors
$= \frac{2x^2}{3y^3}$	simplify

$\frac{x^2 + 5x + 6}{x^2 + 8x + 12}$	Method:
$= \frac{(x+3)(x+2)}{(x+6)(x+2)}$	factor the numerator and the denominator
$= \frac{(x+3)}{(x+6)}$	cancel out common factors

You Try:

1. $\frac{54x^3y^4}{-6xy^5}$

2. $\frac{18x^3y - 9x^2y^2}{3xy}$

3. $\frac{x^2 - x - 2}{x^2 - 3x + 2}$

Adding and Subtracting Algebraic Fractions

To add and subtract algebraic fractions we need to find a common denominator the same way we did with numerical fractions. Once we've added or subtracted we fully simplify

Worked Examples:

$\frac{x}{x+2} + \frac{3}{x+2}$	Method:
$= \frac{x+3}{x+2}$	denominators are the same so simply add the numerators

$\frac{1}{x} - \frac{2}{x+3}$	Method:
$= \frac{1(x+3)}{x(x+3)} - \frac{2x}{x(x+3)}$	the lowest common denominator is $x(x+3)$
$= \frac{1(x+3) - 2x}{x(x+3)}$ $= \frac{x+3-2x}{x(x+3)}$	expand and combine the numerators
$= \frac{3-x}{x(x+3)}$	simplify

$\frac{8x}{x^2 - 16} - \frac{4}{x - 4}$	Method:
$= \frac{8x}{(x - 4)(x + 4)} - \frac{4}{(x - 4)}$	factor the denominators if needed
$= \frac{8x}{(x - 4)(x + 4)} - \frac{4(x + 4)}{(x - 4)(x + 4)}$	rewrite with a common denominator
$= \frac{8x - 4x - 16}{(x - 4)(x + 4)}$ $= \frac{4x - 16}{(x - 4)(x + 4)}$	simplify the numerators
$= \frac{4(x - 4)}{(x - 4)(x + 4)}$	factor the numerator and look for common factors
$= \frac{4}{(x + 4)}$	remove common factors

You Try:

1. $\frac{2}{5x^2y} + \frac{5}{2xy^2}$

2. $\frac{2x}{x^2 - 4} - \frac{1}{x + 2}$

Multiplying Algebraic Fractions

To multiply algebraic fractions we multiply the numerators and we multiply the denominators. We then fully simplify.

Worked Examples:

$\frac{2x}{x^2 - 7x + 12} \cdot \frac{x^2 - 9}{6x^2}$	Method:
$= \frac{2x}{(x-3)(x+4)} \cdot \frac{(x-3)(x+3)}{6x^2}$	factor numerators and denominators
$= \frac{2x(x-3)(x+3)}{6x^2(x-3)(x-4)}$	write the multiplication with the monomials first
$= \frac{(x+3)}{3x(x-4)}$	divide out the common factors

You Try:

1.
$$\frac{5x}{x^2 + 5x + 6} \cdot \frac{x^2 - 4}{10x}$$

2.
$$\frac{3n^2 - 8n - 3}{n^2 - 25} \cdot \frac{n^2 + 10n + 25}{2n^2 - 7n + 3}$$

Dividing Algebraic Fractions

To divide algebraic fractions we rewrite the division as the product of the first rational expression and the reciprocal of the second.

Worked Examples:

$\frac{x^2 - 4x - 12}{x^2 + 8x + 12} \div \frac{x^2 - 36}{3x}$	Method:
$= \frac{x^2 - 4x - 12}{x^2 + 8x + 12} \cdot \frac{3x}{x^2 - 36}$	rewrite as a product
$= \frac{(x - 6)(x + 2)}{(x + 6)(x + 2)} \cdot \frac{3x}{(x - 6)(x + 6)}$	factor numerators and denominators
$= \frac{3x(x - 6)(x + 2)}{(x - 6)(x + 6)(x + 6)(x + 2)}$	write the multiplication with the monomials first
$= \frac{3x}{(x + 6)(x + 6)}$	divide out the common factors
$= \frac{3x}{(x + 6)^2}$	simplify if necessary

You Try:

1. $\frac{11 + x}{x - 9} \div \frac{121 - x^2}{9 - x}$

2. $\frac{\frac{3x^2 - 12x - 63}{4x + 3}}{6x^2 - 42x}$

Rational Equations (Equations Involving Algebraic Fractions)

To solve equations involving algebraic fractions, we will multiply both sides of the equation by the lowest common denominator, giving us an equation that does not contain a rational expression.

An **extraneous solution** to a rational equation is an algebraic solution that would cause any of the expressions in the original equation to be undefined.

Worked Examples:

$1 - \frac{5}{y} = -\frac{6}{y^2}$	Method:
$y \neq 0$	Note any value of the variable that would make any denominator zero.
$LCD = y^2$	Find the least common denominator of all denominators in the equation.
$y^2\left(1 - \frac{5}{y}\right) = y^2\left(-\frac{6}{y^2}\right)$	Clear the fractions by multiplying both sides of the equation by the LCD.
$y^2 - \frac{5y^2}{y} = -\frac{6y^2}{y^2}$ $y^2 - 5y = -6$ $y^2 - 5y + 6 = 0$ $(y - 2)(y - 3) = 0$ $y = 2 \text{ or } y = 3$	Solve the resulting equation.
<i>we did not get $y = 0$ as a solution</i>	if any values are extraneous , discard them

$\frac{n+11}{n^2-5n+4} = \frac{5}{n-4} - \frac{3}{n-1}$	Method:
$\frac{n+11}{(n-4)(n-1)} = \frac{5}{n-4} - \frac{3}{n-1}$ $\text{so } n \neq 4, n \neq 1$	Note any value of the variable that would make any denominator zero.
$LCD = (n-4)(n-1)$	Find the least common denominator of all denominators in the equation.
$(n-4)(n-1)\frac{n+11}{(n-4)(n-1)} = (n-4)(n-1)\frac{5}{n-4} - (n-4)(n-1)\frac{3}{n-1}$	Clear the fractions by multiplying both sides of the equation by the LCD.
$n+11 = (n-1) \cdot 5 - (n-4) \cdot 3$ $n+11 = 5(n-1) - 3(n-4)$ $n+11 = 5n-5-3n+12$ $n = 4$	Solve the resulting equation.
<i>4 is not a valid solution so is discarded</i> <i>$\therefore$ there is no solution to this problem</i>	if any values are extraneous , discard them

You Try:

1. $\frac{2}{x+1} + \frac{1}{x-1} = \frac{1}{x^2-1}$

2. $\frac{y}{y+6} = \frac{72}{y^2-36} + 4$